

Lucien Brey

MACHINE LEARNING CHEMIST

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Working experience

Atinary Technologies Inc.

Lausanne, Switzerland

MACHINE LEARNING SCIENTIST

Nov. 2024 - Now

- Development and maintenance of proprietary Bayesian optimization algorithms (GPs, NPs and generative models)
- Augment optimization scheme with physical and chemical prior
- Collaboration with academic partners such as Prof. J.M Hernandez-Lobato (Cambridge), Prof. Philippe Schwaller (EPFL) and Prof. Angelo Frei (York)

Atinary Technologies Inc.

Lausanne, Switzerland

MACHINE LEARNING AND DATA SCIENCE

Aug. 2023 - Jan. 2024

INTERN

- Developing a NLP-based tool to enable database and literature search from user request for more efficient transfer learning
- Benchmarking and optimization of existing algorithms

EPFL

Lausanne, Switzerland

TEACHING ASSISTANT

- TA in the Advanced General Chemistry course for first year bachelor students in EPFL Life Sciences program and UNIL Geosciences — 2020-2023

Education

École Polytechnique Fédérale de Lausanne (EPFL)

Lausanne, Switzerland

STUDENT

2019-2024

- MSc in Molecular and Biological Chemistry — Sep. 2022-Sep. 2024
- BSc in Chemistry and Chemical Engineering — Sep. 2019-June 2022

Massachusetts Institute of Technology (MIT)

Cambridge MA, USA

MASTER THESIS

Mar. 2024 - Sep. 2024

- *Differentiable inverse design of metal oxides and perovskites catalyst via fine-tuning of surface oxygens site-properties* supervised by Prof. Rafael Gómez-Bombarelli

Kungliga Tekniska Högskolan (KTH)

Stockholm, Sweden

EXCHANGE YEAR

Aug. 2021 - June 2022

- Exchange year abroad, discovering the culture, language and educational system in Sweden

Publications

S. Gallarati, P. van Gerwen, R. Laplaza, **L. Brey**, A.

2024

Makaveev and C. Corminboeuf

A genetic optimization strategy with generality in

asymmetric organocatalysis as a primary target,

Chemical Science, Vol. 15, Issue 10, 3640-3660

F. A. Formica, E. Wu, **L. Brey**, D. Pacheco Gutiérrez,

2025

R. Barbano, H. Tribukait, J. M. Hernández-Lobato, P.

Laveille, L. M. Roch

Bridging Innovation and Efficiency: The Promises and

Challenges of Self-Driving Labs as Sustainable Drivers

for Chemistry, Chimia, Vol. 79, Issue 9, 600-605

Projects

EPFL, Prof. Philippe Schwaller

Lausanne, Switzerland

PROJECT STUDENT

Feb. 2023 - Jul. 2023

- Completing USPTO-MIT database with stoichiometry and reaction conditions, together with by-products, based on reference private database
- Using transformers for reaction completion and byproduct prediction
- Developed an Streamlit web app for reaction prediction

EPFL, Prof. Clémence Corminboeuf

Lausanne, Switzerland

PROJECT STUDENT

Sep. 2022 - Jan. 2023

- Building database based on experimental results for enantioselectivity of catalyst/substrate couples for Pictet-Spengler reaction
- Developing quantum machine learning model to shape a good fitness function capable to predict the key parameter in selectivity for a new catalyst/substrate structure

KTH, Prof. Mårten Ahlquist

Stockholm, Sweden

RESEARCHER ASSISTANT

Jan 2022-Jun. 2022

- Mechanism rationalization about ligand-controlled product selectivity in electrochemical carbon dioxide reduction using manganese bipyridine catalysts using QM Jaguar methods
- Structure optimization, transition state search and 3D modelling on Schrodinger Maestro software

Associations

EPFL Chemistry Student Society (AdEC)

Lausanne, Switzerland

PRESIDENT

Sep. 2022-Aug. 2023

- Organizing events such as Weekend Ski, Fondue de Section, Welcome Day for new students, quiznights in a bar etc.
- Coordination of seminars with the Swiss Chemical Society (SCS), professors, companies and alumni

KTH THS International

Stockholm, Sweden

BUDDY

Jan. 2022-Mar. 2022

- Responsible of a group of international students, organizing, preparing and staffing during events such as museum visits or trivia nights

Coaching Chimie EPFL

Lausanne, Switzerland

TREASURER

Sep. 2020-Jun. 2021

- In charge of the association expenses, expense report, budget etc.
- Organization of events (game nights, picnics...) for the first year students to help them socialize and get introduced to the student life

Skills

Languages French C2 | English C2 | Spanish C1 | Italian A2

Programming (Python) | Bayesian Optimization |
Probabilistic Machine Learning | Generative Modeling

Skills | Reinforcement Learning | Organic Chemistry |
Inorganic Chemistry | Quantum Chemistry |
Molecular Dynamics